

Mock Exam 1 — Solutions

Lectures 1–4 (Chapters 1–3)

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Quantitative Research Methods · HSBI

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Overview of the mock exam

Problem	Topic	Points
P1	Fundamentals of statistical learning	15
P2	Bias–variance decomposition	15
P3	K -nearest neighbours classification	12
P4	Simple linear regression by hand	24
P5	Multiple regression — interpreting Python output	24
Σ		90

Exam conditions

90 minutes for 90 points (≈ 1 point per minute). Allowed aids: non-programmable calculator and one handwritten A4 sheet (both sides). Justify every answer; round final results to 3 decimals.

Problem 1 — task

Problem 1 (15 P) — Fundamentals of statistical learning

- (a) [4 P] Prediction vs. inference — explain the difference; one example each.
- (b) [4 P] Parametric vs. non-parametric estimation of f ; one advantage of each; which flexibility–interpretability trade-off is behind the choice?
- (c) [3 P] Classify: (i) churn prediction for 800 telecom customers; (ii) which firm characteristics are associated with CEO salary (90 firms); (iii) find groups among 5,000 shopping baskets.
- (d) [4 P] Why is training MSE generally smaller than test MSE? What is overfitting?

Problem 1 — solution (a) and (b)

Solution P1 — (a) and (b)

(a) Prediction: accurate $\hat{Y} = \hat{f}(X)$ for new X ; $\hat{f}$ may be a black box. *Example:* is an incoming e-mail spam?

Inference: understand *how* Y depends on $X_1, \dots, X_p$ (which predictors, direction, size); $\hat{f}$ must be interpretable. *Example:* which ad channels increase sales and by how much?

(b) Parametric: (1) assume a form, e.g. $f(X) = \beta_0 + \sum_j \beta_j X_j$; (2) estimate the few parameters (least squares). *Advantage:* needs less data; simple and interpretable.

Non-parametric: no functional form; $\hat{f}$ follows the data (KNN, splines). *Advantage:* captures many possible shapes of f — no risk of a badly wrong form.

Trade-off: flexibility reduces bias but costs data, risks overfitting and loses interpretability; for inference we usually prefer restrictive models.

Problem 1 — solution (c) and (d)

Solution P1 — (c) and (d)

(c) (i) *Classification*: qualitative response cancel yes/no (prediction).

(ii) *Regression*: quantitative response CEO salary (inference).

(iii) *Unsupervised*: no response variable; goal is clustering.

(d) The method is fitted by minimising *training* error, so $\hat{f}$ adapts to the training observations *including their noise* — training MSE systematically underestimates the error on fresh data. With growing flexibility the training MSE falls monotonically while the test MSE is U-shaped.

Overfitting: the fit follows patterns of the training noise that do not exist in the population; a less flexible model would have *lower test* MSE. Symptom: very small training MSE together with a rising test MSE (widening gap).

Problem 2 — task

Problem 2 (15 P) — Bias–variance decomposition

At a fixed x_0 three methods were analysed; $\sigma^2 = \text{Var}(\varepsilon) = 1.00$:

	A (linear)	B (moderate)	C (very flexible)
$[\text{Bias}]^2$	2.25	0.16	0.01
$\text{Var}(\hat{f}(x_0))$	0.25	1.44	3.60

(a) **[4 P]** State the decomposition of the expected test MSE and explain the components. (b) **[5 P]** Compute the expected test MSE for A–C; which method to choose? (c) **[3 P]** Smallest expected test MSE any method could achieve here? (d) **[3 P]** Behaviour of bias²/variance/training/test MSE as flexibility grows; U-shape.

Problem 2 — solution (a) and (b)

Solution P2 — (a) and (b)

$$(a) \quad E[(y_0 - \hat{f}(x_0))^2] = \underbrace{\text{Var}(\hat{f}(x_0))}_{\text{sensitivity to training set}} + \underbrace{[\text{Bias}(\hat{f}(x_0))]^2}_{\text{systematic model error}} + \underbrace{\text{Var}(\varepsilon)}_{\text{irreducible noise}}$$

(b) Expected test MSE = Bias² + Var + σ^2 :

	Bias ²	Var	σ^2	Test MSE
A	2.25	0.25	1.00	3.50
B	0.16	1.44	1.00	2.60
C	0.01	3.60	1.00	4.61

Choose **B**: lowest expected test MSE. C pays more in variance (+2.16) than it gains in bias (−0.15).

Problem 2 — solution (c) and (d)

Solution P2 — (c) and (d)

(c) $\sigma^2 = 1.00$. Since $\text{Bias}^2 \geq 0$ and $\text{Var} \geq 0$, $E[(y_0 - \hat{f}(x_0))^2] \geq \text{Var}(\varepsilon) = 1.00$ for every method. Even the true f cannot predict the noise draw ε_0 : the *irreducible error* is the floor.

(d) As flexibility increases:

- Bias^2 *decreases* (typically monotonically); variance *increases*;
- training MSE decreases *monotonically* (the fit hugs the training data);
- test MSE = $\text{Bias}^2 + \text{Var} + \sigma^2$ first falls (bias reduction dominates) then rises (variance dominates) $\Rightarrow$ **U-shape**.

In the table: $3.50 \rightarrow 2.60 \rightarrow 4.61$ for $A \rightarrow B \rightarrow C$ — minimum at intermediate flexibility.

Problem 3 — task

Problem 3 (12 P) — KNN classification

Seven visitors with scores X_1 (pages) and X_2 (minutes) and outcome Y (purchase); new visitor at $x_0 = (4,2)$ — classify with Euclidean distance.

Obs.	1	2	3	4	5	6	7
X_1	4	2	6	4	1	7	5
X_2	3	2	2	5	1	4	6
Y	Yes	No	No	No	Yes	Yes	No

- (a) **[4 P]** All distances; classification with $K = 1$. (b) **[4 P]** Classification with $K = 3$ and $\hat{Pr}(\text{Yes} \mid x_0)$.
(c) **[2 P]** Effect of K on bias/variance; training error of 1-NN. (d) **[2 P]** Bayes classifier: definition; why unattainable; relation to KNN.

Problem 3 — solution (a) and (b)

Solution P3 — (a) and (b)

(a) $d^2 = (X_1 - 4)^2 + (X_2 - 2)^2$, sorted:

Obs.	1	2	3	4	5	6	7
d^2	1	4	4	9	10	13	17
d	1.000	2.000	2.000	3.000	3.162	3.606	4.123
Y	Yes	No	No	No	Yes	Yes	No

$K = 1$: nearest is Obs. 1 ($d = 1.000$; Yes) $\Rightarrow$ **Yes**.

(b) $K = 3$: neighbours are Obs. 1 (Yes) and Obs. 2 and Obs. 3 (both No). $\hat{\Pr}(\text{Yes} \mid x_0) = 1/3 \approx 0.333$; $\hat{\Pr}(\text{No} \mid x_0) = 2/3 \Rightarrow$ majority vote: **No**. Note that $K = 1$ and $K = 3$ disagree — the answer depends on the flexibility.

Problem 3 — solution (c) and (d)

Solution P3 — (c) and (d)

(c) Small K (e.g. 1): highly flexible and jagged boundary $\Rightarrow$ *low bias, high variance*. Large K : strong local averaging, smooth almost linear boundary $\Rightarrow$ *high bias, low variance*. Training error of 1-NN: 0 — each training point is its own nearest neighbour (says nothing about test error).

(d) The *Bayes classifier* assigns x to the class j maximising the *true* conditional probability $\Pr(Y = j \mid X = x)$; it minimises the expected test error (*Bayes error rate* = lower bound for all classifiers). Unattainable in practice: the conditional distribution of Y given X is unknown. KNN *estimates* these probabilities locally from the K nearest neighbours and then classifies to the largest estimate — a data-driven approximation of the Bayes rule.

Problem 4 — task

Problem 4 (24 P) — Simple linear regression by hand

Radio spots x and weekly sales y (in 1,000 EUR) for $n = 5$ weeks; $\sum x_i = 15$; $\sum y_i = 55$; $\sum x_i^2 = 55$; $\sum x_i y_i = 195$.

Week	1	2	3	4	5
x_i	1	2	3	4	5
y_i	6	7	11	13	18

- (a) [6 P] $\hat{\beta}_1$ and $\hat{\beta}_0$; fitted line; interpretation. (b) [4 P] Fitted values and residuals; check $\sum e_i = 0$. (c) [3 P] RSS and RSE; interpretation. (d) [6 P] $SE(\hat{\beta}_1)$; t -test of $H_0: \beta_1 = 0$ ($t_{0.975,3} = 3.182$); 95 % CI. (e) [3 P] TSS and R^2 ; interpretation. (f) [2 P] Prediction at $x = 6$; one caveat.

Problem 4 — solution (a) and (b)

Solution P4 — (a) and (b)

(a) $\bar{x} = 3$; $\bar{y} = 11$; $S_{xy} = 195 - 5 \cdot 3 \cdot 11 = 30$; $S_{xx} = 55 - 5 \cdot 3^2 = 10$

$$\hat{\beta}_1 = \frac{30}{10} = 3.000 \quad \hat{\beta}_0 = 11 - 3 \cdot 3 = 2.000 \quad \hat{y} = 2.000 + 3.000x$$

Slope: +1 spot $\Rightarrow$ expected sales +3,000 EUR per week. *Intercept*: expected sales without spots = 2,000 EUR (mild extrapolation; $x = 0$ not observed).

(b) $\hat{y}_i = 2 + 3x_i$; $e_i = y_i - \hat{y}_i$:

i	1	2	3	4	5
$\hat{y}_i$	5	8	11	14	17
e_i	1	-1	0	-1	1

$\sum e_i = 1 - 1 + 0 - 1 + 1 = 0 \checkmark$ (always true for least squares with intercept).

Problem 4 — solution (c) and (d)

Solution P4 — (c) and (d)

(c) $RSS = 1 + 1 + 0 + 1 + 1 = 4.000$; $RSE = \sqrt{4/(5-2)} = \sqrt{4/3} = 1.155$ (thousand EUR): typical deviation of observed sales from the true line ($\hat{\sigma}$).

(d) $SE(\hat{\beta}_1) = \frac{RSE}{\sqrt{S_{xx}}} = \frac{1.1547}{\sqrt{10}} = 0.365$; $t = \frac{3.000}{0.3651} = 8.216$ (df = 3).

- $|t| = 8.216 > 3.182 = t_{0.975,3} \Rightarrow$ reject $H_0: \beta_1 = 0$ at the 5 % level: significant positive association.
- 95 % CI: $3.000 \pm 3.182 \cdot 0.3651 = 3.000 \pm 1.162 = [1.838; 4.162]$; $0 \notin$ CI — consistent with the test.

Problem 4 — solution (e) and (f)

Solution P4 — (e) and (f)

(e) $TSS = \sum_i (y_i - 11)^2 = 25 + 16 + 0 + 4 + 49 = 94.000$

$$R^2 = 1 - \frac{RSS}{TSS} = 1 - \frac{4}{94} = 0.957$$

95.7 % of the variance of weekly sales is explained by the regression on radio spots (simple regression: $r = \sqrt{R^2} \approx 0.979$).

(f) $\hat{y}(6) = 2 + 3 \cdot 6 = 20.000$ thousand EUR = 20,000 EUR.

Caveat: $x = 6$ lies outside the observed range $[1;5] \Rightarrow$ **extrapolation** — the linear form is unverified there (and $n = 5$ is very small).

Problem 5 — task

Problem 5 (24 P) — Multiple regression output

$n = 60$ employees; salary (1,000 EUR/year); experience (years); female (dummy). OLS fit of salary = $\beta_0 + \beta_1 \text{experience} + \beta_2 \text{female} + \beta_3 (\text{experience} \times \text{female}) + \varepsilon$:

	coef	std err	t	$P > t $
Intercept	46.2000	1.100	42.000	<0.0001
experience	2.3500	0.250	9.400	<0.0001
female	-3.1000	1.550	-2.000	0.0504
experience:female	-0.8500	0.350	-2.429	0.0184

$R^2 = 0.712$; adj. $R^2 = 0.697$; $F = 46.15$ on 3 and 56 df; $\text{Prob}(F) < 0.0001$

(a) [6 P] Interpret all coefficients. (b) [4 P] Significance at 5 %; drop female? (c) [4 P] Fitted equation per group. (d) [4 P] Predictions at 10 years; gap. (e) [3 P] R^2 and F -test. (f) [3 P] U-shaped residual plot; VIF = 12.

Problem 5 — solution (a)

Solution P5 — (a) coefficient interpretation

Salary in 1,000 EUR per year; baseline group: men (`female=0`).

- **Intercept 46.20:** expected salary of a *man* with 0 years of experience = 46,200 EUR.
- **experience 2.35:** for *men* each additional year raises expected salary by 2,350 EUR (with an interaction the slope is group-specific).
- **female -3.10:** at 0 years of experience women earn 3,100 EUR *less* on average — the difference of the two intercepts.
- **experience:female -0.85:** the experience slope of women is 0.85 smaller: women gain $2.35 - 0.85 = 1.50$ thousand EUR per year; the expected gender gap *widens* by 850 EUR per year of experience.

Problem 5 — solution (b) and (c)

Solution P5 — (b) and (c)

(b) Intercept and experience: $p < 0.0001 \Rightarrow$ significant. Interaction: consistency $t = -0.85/0.35 = -2.429 \checkmark$; $p = 0.0184 < 0.05 \Rightarrow$ significant. female: $p = 0.0504 > 0.05 \Rightarrow$ *not* significant at exactly 5 % (borderline). Do *not* remove it: **hierarchy principle** — a model containing an interaction should contain the corresponding main effects; and $p = 0.0504$ is weak evidence against an effect, not proof of none.

(c) Group-wise fitted equations:

$$\text{Men: } \widehat{\text{salary}} = 46.20 + 2.35 \text{ experience}$$

$$\text{Women: } \widehat{\text{salary}} = (46.20 - 3.10) + (2.35 - 0.85) \text{ experience} = 43.10 + 1.50 \text{ experience}$$

Different intercepts *and* slopes — exactly what the interaction permits.

Problem 5 — solution (d) and (e)

Solution P5 — (d) and (e)

(d) At 10 years of experience:

$$\text{Man: } 46.20 + 2.35 \cdot 10 = 69.70 \Rightarrow 69,700 \text{ EUR}$$

$$\text{Woman: } 43.10 + 1.50 \cdot 10 = 58.10 \Rightarrow 58,100 \text{ EUR}$$

$$\text{Gap: } 58.10 - 69.70 = -11.60 \text{ thousand EUR} = \hat{\beta}_2 + 10\hat{\beta}_3 = -3.10 - 8.50 \Rightarrow \text{women earn 11,600 EUR less.}$$

(e) $R^2 = 0.712$: the three terms explain 71.2 % of the salary variance. F -test: $H_0: \beta_1 = \beta_2 = \beta_3 = 0$ (no predictor useful) vs. at least one $\beta_j \neq 0$; $F = 46.15$ (3 and 56 df) with $p < 0.0001 \Rightarrow$ reject H_0 : the model is useful overall.

Problem 5 — solution (f) diagnostics

Solution P5 — (f) diagnostics

(i) **U-shaped residual-vs-fitted plot:** systematic curvature the linear model misses $\Rightarrow$ **non-linearity**. Remedies: add a polynomial term (e.g. experience^2) or transform variables (e.g. $\log(\text{salary})$ or $\sqrt{X}$).

(ii) **VIF = 12 for age:** $\text{VIF}_j = 1/(1 - R_{X_j|X_{-j}}^2) = 12 \Rightarrow R_{X_j|X_{-j}}^2 \approx 0.917$: age is almost a linear combination of the other predictors $\Rightarrow$ strong **multicollinearity** (rules of thumb: $\text{VIF} > 5$ or > 10 is problematic).

Consequence: $\text{SE}(\hat{\beta}_{\text{age}})$ inflated by $\sqrt{12} \approx 3.5 \Rightarrow$ unstable coefficient; low power of its t -test; individual effects of collinear predictors cannot be separated — although overall fit and predictions may remain fine.